InpaintAR

A Comparison of different InPainting Approaches for Real-Time Use

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Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere. This printed copy is identical to the submitted electronic version.

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Contents

Abstract

Due to technological advances in both Hardware and Software-Integration, Augmented Reality is getting more popular with each year.
One of the issues still prevalent in Augmented Reality Software is the distraction caused by clutter surrounding the digital content in use cases like immersive analytics.

This bachelors thesis focuses on the analysis of the different methods and algorithms for the concealment of the clutter to provide a clean and non-distracting environment for the digital object to be displayed on.
To accomplish this, a tool to select planes on which objects are detected to subsequently be hidden through inpainting will be implemented. With this tool, several methods and algorithms will be evaluated in terms of both quality and performance to provide believable and uncluttered environments, whilst at the same time minimizing the use of computational resources and time.
These findings are compared to each other so readers can get an overview on both the advantages and disadvantages each of the analyzed approaches delivers.

Kurzfassung

Aufgrund der technologischen Fortschritte, sowohl in der Hardware, als auch in der Software-Integration, steigt die Beliebtheit von Augmented Reality mit jedem Jahr an. Ein sehr gängiges Problem in aktueller Augmented Reality Software ist die Ablenkung, welche durch physische Objekte, die um den digitalen Inhalten liegen. Beispielsweise bei der immersiven Analyse von Daten.

In dieser Bachelorarbeit liegt der Fokus auf der Analyse verschiedener Methoden und Algorithmen zur Verdeckung dieser Objekte, wodurch eine aufgeräumte, saubere Umgebung für die digitalen Elemente geschaffen wird.
Um dies zu erreichen wird ein Werkzeug entwickelt, welches erlaubt Flächen zu wählen, welche dann dazu verwendet werden, um Objekte auf diesen Flächen zu finden und darauffolgend durch die Nutzung von Inpainting zu verdecken. Mithilfe dieses Werkzeugs werden mehrere Methoden und Algorithmen sowohl auf Qualität, als auch auf Leistung evaluiert, um sowohl die Glaubbarkeit und Ordnung der generierten Umgebungen, während parallel dazu ein Fokus auf die Minimierung von Rechenresourcen und Laufzeit gelegt wird.
Diese Ergebnisse werden zueinander verglichen, sodass Leser sowohl über die Vor- als auch Nachteile der jeweiligen Ansätze einen Überblick bekommen.

Chapter 1

Introduction

The domain of Augmented Reality (AR) spans all kinds of real-time visualization approaches wherein an otherwise real environment is augmented with virtual objects [Mil+95] While many still considerAR a nieche, the user-numbers tell a different story. In 2023 Statista reported about 983 Million users to be using AR on their mobile devices. [Sta24].

One of the issues still common in AR Applications is the overload of information. Whenever users are supposed to focus on the virtual objects at hand, the visibility of physical objects may serve as a distraction. (cf. Cheng et. al. [Che+22]) An existing solution is the use of Diminished Reality (DR). This concept can be interpreted as a specific usecase of AR, which aims to hide or remove certain physical objects from the viewport in real time.[MF02] (see 1.1). In the case of the tool presented in this thesis, DR can be used to reduce the visual clutter.

To achieve this the visual clutter from the viewport is intentionally drawn over, artificially damaging the frame shown to the user on the display. To fix such damaged images, image inpainting is one of the techniques available and the one used in this paper.

TODO

Comparison (a) real image, (b) Image with Mask, (c) Mask Inpainted

Figure 1.1: Real and Diminished scene. This example serves as an example of how DR may be used to reduce visual clutter. (a) shows the regular environment, (b) shows the same image with the visual clutter overdrawn, whilst (c) shows the cleaned up DR image, achieved through inpainting

1.1 Motivation

TODO ADD SMALL PART ABOUT VISUAL CLUTTER

While there are existing solutions for achieving DR using inpainting, see [Thi+24], the topic of which inpainting method for not only this usecase, but inpainting in a 3-Dimensional Real-Time Environment has largely been disregarded as of now. Most publications put the focus on different components of the system like the detection of the objects, user studies or optimizations of smaller visual artifacts. Over the last years there have been numerous achievements in the space of inpainting methods. Ranging from classic, low-cost algorithms like the fast marching algorithm [SHD21] or the usage of matching textures using gaussian weighting to fill the affected regions [DRR19] to more modern machine-learning assisted high fidelity approaches like the DeepFill Deep-Learning model [CJL23].

The main goal of this analysis is to provide an easy to reference comparison of the different methodologies to be compared in terms of both performance and quality. Through this analysis, developers can make an educated choice on which methodology to use for their usecase, be it high fidelity in usecases like PC-powered AR systems or low energy, low performance systems, more suited for mobile devices, be they mobile phones or standalone Mixed-Reality Head-Mounted Displays (HMDs). In addition, the tool built for this evaluation should be easy to extend for newer inpainting methods whilst staying easy to integrate into existing applications to allow for a quick and efficient tool to provide the clutter removal functionality to other AR projects.

1.2 Overview

This work includes an introduction to the underlying concepts and analyzed inpainting methods, followed by the conceptualization and implementation of the tool as well as the performance and quality evaluation for each of the analyzed inpainting methods. These topics are elaborated upon in the following five chapters:

- **Chapter 2: Fundamentals and Related Work**
  At the start of the thesis, this chapter serves to provide an introduction into visual clutter and why it matters to the domain of Augmented Reality. The term of DR is explored into further detail as well as the underlying logic of each of the analyzed inpainting methods.
- **Chapter 3: Concept**
  The concept chapter goes into further detail on the architecture and design chosen for this tool and provides an overview on a few example use cases for which the tool might be used.
- **Chapter 4: Implementation**
  In this chapter each of the classes contained in this tool is explained briefly. In addition a small guide is being provided, as to how the tool could be implemented for the usage in existing projects. Some of the tools and principles used during the implementation are going to be elaborated on as well.
- **Chapter 5: Evaluation**

After providing an overview of the testing setup and how visual clutter and performance are measured, the test scenarios will be presented, followed by the results for each of the analyzed methods.

- **Chapter 6: Conclusion and Future Work**

  At the end of the thesis all the results and the everything mentioned in the previous chapters is briefly summed up. Afterwards further ideas on how to expand and improve the tool in the future are provided to improve both the developer and user experience even further.

Chapter 2

Fundamentals and Related Work

2.1 What is Visual Clutter

2.2 Why does it matter

Use Distractor Suppression Paper as reference

2.3 Diminished Reality

2.4 Inpainting

2.4.1 What is inpainting

2.4.2 Methods analyzed in this thesis

Fast Marching Method

Texture Synthesis and Region Segmentation

Nonlocal Texture Matching and nonlinear filtering

Still TBD, depends if too similar to Texture Synthesis or not

Nearest Neighbor Pixel Filling

DeepLearning

Regression

2.4.3 Inpainting in Diminished Reality

Chapter 3

Concept

Go into Detail about the overview, how it is built architecturally, what are the components, explained in further sections

3.1 Components

3.2 Requirements

Small explanation of what requirements exist, unity

3.2.1 Hardware

go into detail about meta quest 3

3.2.2 Meta AR Library

Explain the AR Library, how it is used / why needed

3.2.3 Unity XR Hand-Tracking Library

Explain Gestures / Hand Pose Recognition

3.3 Usecases

Go into Detail about how the tool might prove useful

3.3.1 Interactive Analysis

Maybe ask canthes if i can reference some of the interactive analysis projects from HIVE as references / images

3.3.2 AR Training Simulations

3.3.3 Entertainment Purposes

TODO add Image with virtual Chessboard on cluttered desk

Chapter 4

Implementation

Explain how it is implemented

4.1 Simulating the Headset for development

Explain how the Meta XR Simulator was used during implementation to allow for faster development without needing the headset as much

4.2 Area Selection

Go into Detail on how Hand Gesture Rectangle is evaluated to area and added as mask on top, how the mask is gotten from the area through snakes

4.3 Inpainting

Further Details about the Inpainting Algorithms, per Algorithm

4.4 Mask Overlay

How the inpainted content is put in front of the HMD Camera feed without getting into the way of AR elements

Chapter 5

Evaluation

When the Evaluation was conducted, in what environment

5.1 Testing Setup

5.1.1 Hardware Setup

1x Meta Quest 3 Standalone, 1x Desktop VR with dedicated GPU through Meta Quest 3 with USB-C

Maybe show headset setup, probably Ladder with Headset strapped to something so it can be tested without user input / movement

5.1.2 Measuring Quality for Inpainting

Quality Benchmarks

Use InPainting Quality Benchmarking, Part with Quality Benchmarking without knowing what it should look like, also mention that as a small issue in the evaluation

How visual Clutter is measured

See Paper Measuring Visual Clutter

5.1.3 How Performance is measured

Use Clock/Timer for the Inpainting Method from Call to Return + FrameTime + FrameRate Maybe use the Unity3D Profiler

5.2 Test Scenarios

5.2.1 Table with Pen and Notebook

as said in subsection + Visual Clutter Metric

5.2.2 Table with reflective see-through objects

Table with multiple glasses of water + Visual Clutter Metric

5.2.3 Heavily Cluttered Desk

Desk full with different kinds of junk, clean up all elements on top + Visual Clutter Metric

5.2.4 Large Objects

Probably entire Table / Desk + Visual Clutter Metric

5.3 Results

5.3.1 Quality Metrics

SubSubsection for each of the methods evaluated Graph of the quality metrics (Quality + Visual Clutter remaining) per test-scenario for each of the methods + explanation

5.3.2 Performance Metrics

SubSubsection for each of the methods evaluated Graph of the performance metrics per test-scenario for each of the methods + explanation

Chapter 6

Conclusion and Future Work

Still TODO

6.1 Future Work

6.1.1 Application to selective planes in VR environments

6.1.2 Combination with Dynamic Object Detection

SimpSON Paper, Single-Click-Distraction-Detection; Maybe mention RoboDK AI Model for Detecting Objects by Name and providing the user a table to select which objects should be hidden and shown

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